Agentic Travel Planning: Where AI Helped, and Where It Did Not

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Abstract

Many agentic tasks begin with an open-ended goal but are useful only when hard constraints are met. Multi-day travel planning is a clean instance. We describe our entry to the ChinaTravel challenge: a deterministic, environment-grounded planner with an acceptance gate that can only emit itineraries confirmed valid against the environment rules and APIs. AI agents were most valuable in building and improving that system, and selectively useful for language understanding; they did not replace deterministic optimization and verification. The first official held-out submission sharpened this division of labor: grounding generalized, while constraint extraction tuned to one phrasing style did not.

1 Introduction

A growing class of useful tasks asks a system to turn a loosely stated human goal into an output that is not merely plausible but correct. Multi-day travel planning combines fuzzy requirements with exact feasibility. The IJCAI–ECAI 2026 ChinaTravel challenge[1] extends the benchmark of Shao et al. [Shao *et al.*, 2024]: an agent must produce a preference-aligned itinerary over a sandbox of Chinese cities. A request might say:

> "Plan three days in Suzhou, visit the Humble Administrator's Garden between 9 and 11, stay near the old town, avoid taxis, and spend at most CNY 3,000."

The output is structured data, not free text. Hotels, attractions, and restaurants must exist in the sandbox; opening hours, transportation, and costs must be consistent; logical requirements must hold; and commonsense rules such as meal counts must pass. The evaluator measures environmental validity (EPR), logical satisfaction (C-LPR and FPR), and preferences. A single failed hard requirement zeros all preference terms.

The competition has two phases. In the offline first phase, released verifier fields and a generous time budget allowed us to satisfy almost all constraints and then polish preferences. The harder agent phase studied here exposes only query text and environment tools and gives the complete 100-query run one five-hour budget. The agent must infer what to satisfy as well as how to satisfy it.

Our system follows a propose–verify–repair pattern related to LLM-Modulo systems [Kambhampati *et al.*, 2024], but assigns roles asymmetrically: models may interpret language or propose changes; environment APIs establish facts; and deterministic optimization and checks decide what may be emitted.

2 System Design

2.1 Pipeline and trust boundary

Figure 1 follows the example from language to a simplified itinerary. Oracle verifier fields are removed before execution. The only inputs are visible query text and allowed environment tools. Every POI, hotel, restaurant, price, opening hour, and transportation record is read from `WorldEnv`; nothing is generated as a plausible substitute.

2.2 Parsing and normalization

The parser extracts dates, cities, budgets, excluded transportation, named POIs, timed visits, hotel properties, cuisines, and attraction categories. Alternatives remain separate solver variants rather than becoming an impossible conjunction. Normalization resolves names and concepts against each target city's catalog.

Deterministic extraction is fast and testable, but it is not synonymous with understanding. We added phrasing-robust fallbacks for the largest measured recall gaps and an optional Qwen extractor that may fill a missing family but never overwrite a deterministic value. Model-proposed entities and categories are matched against the environment vocabulary before use.

Precision is equally important. Our audit found that in 23–50% of free-prose queries, depending on the set, the parser invented a required POI name that did not exist in the target city. A new default-off catalog check removes only provably unsatisfiable required names and category values. It reduced this measured failure class to zero on the two local sets, but its end-to-end gate was pending at writing time, so we claim no score gain for it.

[1] https://github.com/LAMDA-NeSy/ChinaTravel

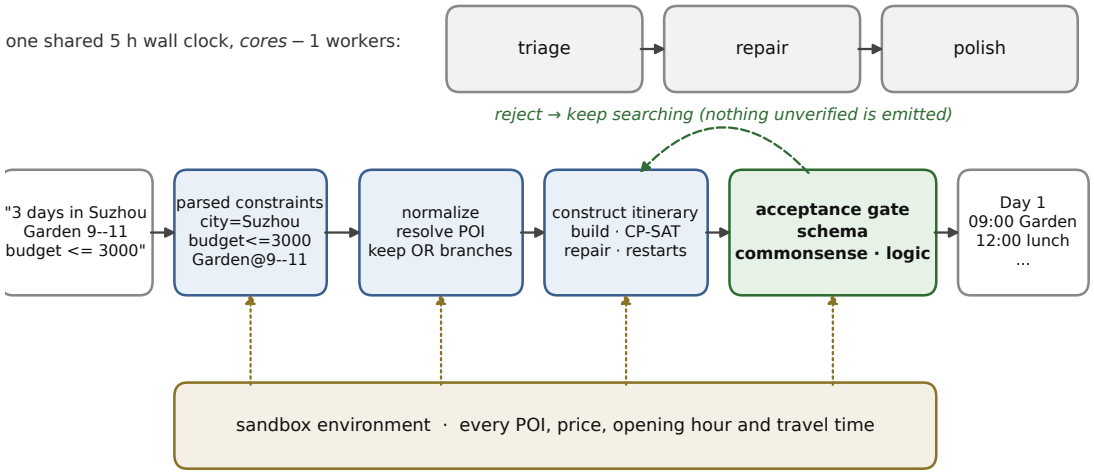


Figure 1: The per-query pipeline and whole-run scheduler, illustrated with the opening example. Optional model-based extraction may fill missing constraints; grounding, construction, repair, and acceptance remain deterministic.

2.3 Construction, repair, and densification

The deterministic planner constructs a complete base itinerary from grounded candidate pools. It then applies bounded CP-SAT [Perron and Didier, 2024], domain-specific repairs, and seeded restarts. Repairs include placing named activities inside requested windows, inserting absent required POIs, closing transportation gaps, and rebuilding intercity legs. The same acceptance checker evaluates every candidate: query constraints, schema, and commonsense rules must all pass. Accepted candidates are ranked by preference value and cost.

Packing and densification add meals and activities only while the checker continues to pass. A minimal schema-valid itinerary is written before solving so an interrupted run still has an output, but this is availability rather than success: an empty schema-valid itinerary scores zero.

2.4 Scheduling the five-hour run

The whole-split scheduler uses *triage*, *repair*, and *polish* passes in parallel across queries. Triage reaches every query cheaply; repair revisits the nearest failures with larger budgets; polish spends remaining time only after hard requirements pass. A keep-best rule makes updates monotonic, and deadline checks prevent starting work that cannot finish within the shared five-hour envelope.

3 Evaluation

3.1 Official feedback

The organizers permit two submitted harness versions and retain the better score. We submitted v1 early to obtain the only available held-out feedback before choosing v2. Table 1 compares that result with the current-main run on 100 familiar released queries. The columns are diagnostic signals, not identically distributed test sets.

Metric	Familiar 100	Held-out v1
Overall	64.17	33.25
EPR micro	99.67	99.48
EPR macro	98.00	95.00
C-LPR	90.08	55.20
FPR	44.00	0.00

Table 1: Current-main local baseline and official v1 result.

Grounding generalized: held-out micro EPR remained 99.48. Constraint understanding did not: C-LPR fell to 55.20, no query passed every requirement, and all preference terms were therefore zero. If the held-out queries have roughly the familiar set's eleven clauses each, the FPR outcome follows naturally from the clause-level rate rather than indicating a second grounding failure.

3.2 Query phrasing

The familiar released set was presented as in distribution, so prioritizing it was a reasonable development choice. Its queries use a regular numbered template, whereas other released sets express similar requirements in varied prose. Offline oracle-backed audits found no missed clauses on the familiar register but 12–29% on the differently phrased sets. Targeted recall fixes raised a matched free-prose score from 41.62 to 49.99 while leaving the familiar control unchanged. The precision audit described in Section 2 then revealed the unsatisfiable-name problem, which can pin an otherwise solvable query at FPR zero regardless of search time.

3.3 Planning baselines

We ran two small, matched comparisons to identify failure modes. An autonomous tool-using Qwen agent scored 1.88 on eight familiar queries and 2.64 on eight free-prose queries, versus 67.71 and 48.49 for our solver; none of its itineraries passed all commonsense checks. The upstream RuleNeSy

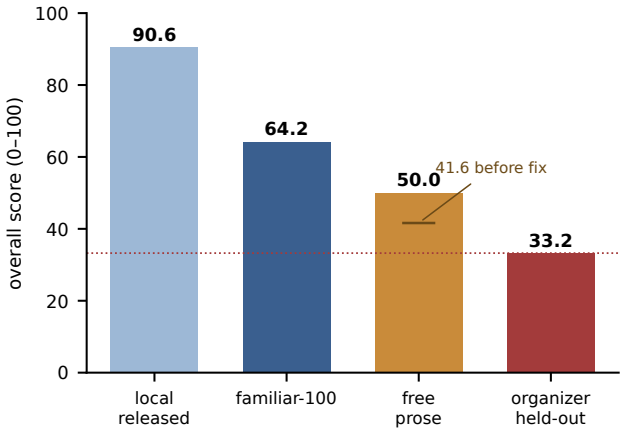


Figure 2: Score changes with phrasing register. "Free prose + fix" is a matched 30-query A/B; held-out is the organizers' v1 score.

baseline [Shao *et al.*, 2024] scored about 9.5 on four queries from each register. It grounded most individual facts but still violated at least one commonsense rule in every itinerary. These samples are too small for general system ranking, but the observed failure modes support retaining deterministic acceptance and optimization.

4 Where AI Helped — and Where It Did Not

Coding agents were force multipliers once we gave them rigorous feedback. Strong agents developed architecture and diagnoses; faster agents implemented focused operators, tests, and evaluations. A deterministic evaluator, targeted regression tests, pinned A/B contracts, and reproducible result ledgers let us reject attractive but invalid changes quickly. AI contributed substantially to the engineering loop without becoming the authority inside the execution loop.

Natural-language interpretation is a more nuanced case. LLM-first extraction cut missing free-prose constraint families by 40–65%, but initially converted many misses into values that did not match the environment vocabulary. We therefore connected model output to the same city-specific entity and category catalogs used by the solver. Grounded, fill-only extraction improved one small familiar A/B, but was flat on the matched free-prose subset (49.99 to 49.88) and still matched fewer clauses than the contemporaneous deterministic parser. The current default consequently makes zero LLM calls, while retaining the grounded extractor as an optional path. This is not a rejection of LLM parsing; it separates recognizing a requirement from resolving its values against the actual database. An improved agent should query those vocabularies directly, whether through the existing tools or an MCP-style interface.

End-to-end itinerary generation was the clearer boundary. Even with environment tools, autonomous Qwen planning accumulated small inconsistencies, and one violation invalidated the whole itinerary. Better prompts and tool interfaces may improve grounding, but dates, opening windows, travel times, budgets, meals, and disjunctions remain a constrained optimization problem. Deterministic solving provides reproducibility, explicit acceptance, and efficient targeted repair.

5 Conclusion

ChinaTravel rewarded a hybrid division of labor. AI agents accelerated the construction and analysis of the system, and model-based extraction recognized language missed by deterministic patterns. Environment tools supplied facts; deterministic search assembled them; and the acceptance gate ensured that only verified itineraries could be emitted. The official v1 result sharpened that boundary: grounding transferred to hidden data, while constraint extraction tuned to one advertised in-distribution phrasing register did not.

The transferable lesson is methodological. Agentic coding works particularly well in a small, isolated project once objective and fast feedback exists. Models are valuable at the fuzzy boundary between language and symbols, provided their proposals are grounded. Hard itinerary construction remains optimization: stochastic fluency does not substitute for satisfying every constraint.

A production system would replace the sandbox with live map, hotel, rail, flight, and booking APIs or MCP servers. It would also infer user-specific objectives rather than inherit fixed challenge preferences: some travelers want four attractions per day, while others value unplanned time. Live availability and uncertain travel times would require replanning, but the same architecture remains useful. Let AI interpret intent and help build the system; let APIs establish reality; let optimization construct the itinerary; and let deterministic verification decide what is safe to emit.

Acknowledgments

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References

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